

Lecture 3

Describing and Visualizing Distributions

Review

1 What do we use data for?

To answer questions about a population.

2 What is a population?

The set of all observations of interest – observed and unobserved.

3 What is a sample?

The subset of the population that we actually observe.

4 What do we call the way we collect the data?

The sampling design.

Review

1 What are the three arms of statistics?

Design, description and inference.

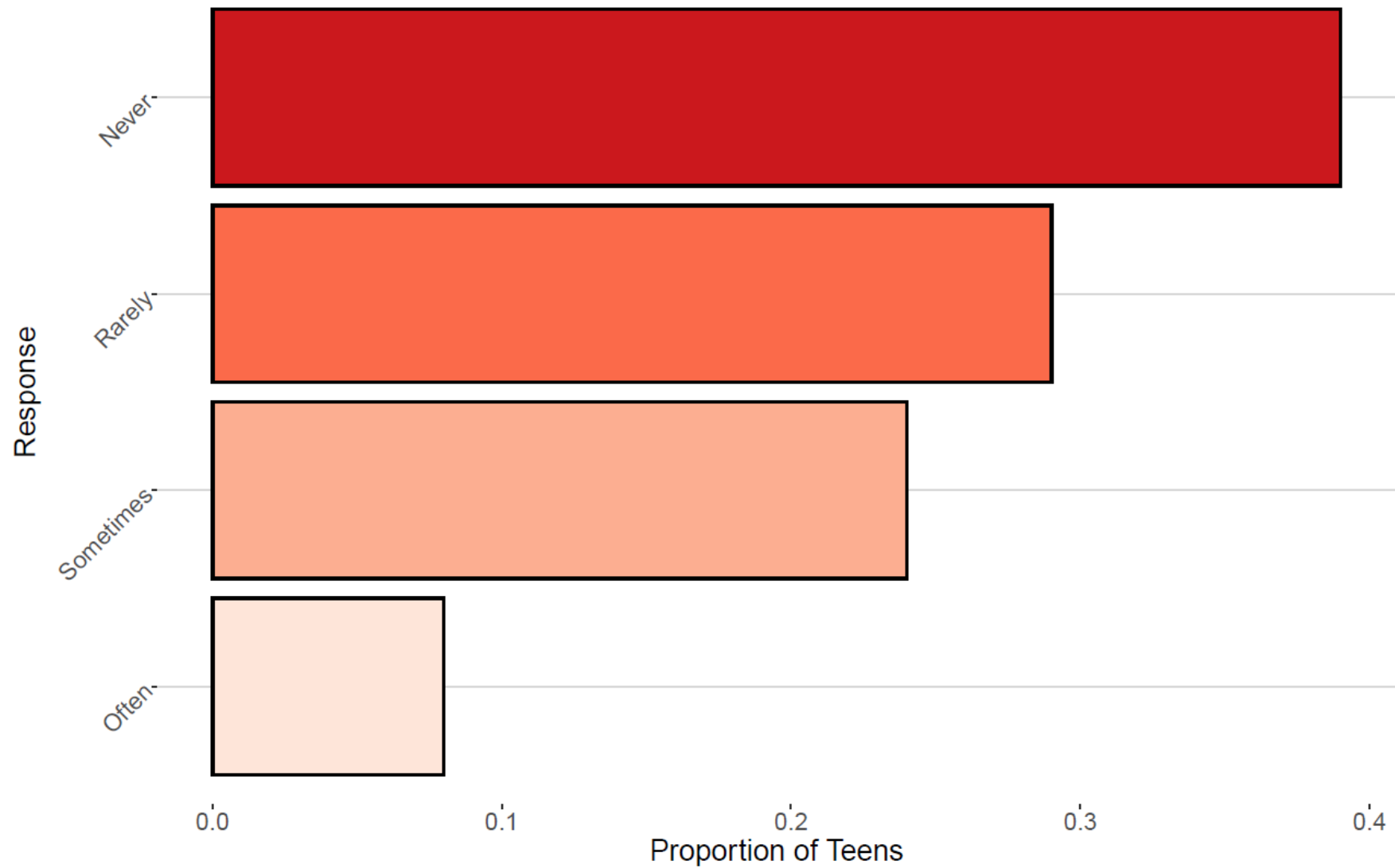
2 Which one comes first, and why?

Design – how we plan to collect the data, settled before any data exists. No amount of description or inference repairs a bad design.

3 What is an estimate?

A statistic computed from a sample, used to approximate an unknown population parameter – together with a statement of how far off it is likely to be.

Are You Losing Focus In Class By Checking Your Cell Phone?





Features of Distributions

- A **distribution** of a variable gives (a) the values that occur and (b) how often each value occurs

Features of A Distribution

Categorical Variables:

- **Modal category** – the category with the highest frequency

Quantitative Variables:

- **Shape** – do observations cluster into certain areas, are values spread out or more densely packed together?
- **Center** – where is the middle point on the distribution, where does a typical value fall?
- **Variability** – how tightly to observations cluster around the center of the distribution

Displaying Distributions: Frequency Tables

- **Frequency table** – a table listing the distinct values of variable together with the number of observations of each value
- **Frequency** – the number of times a value occurs
- **Relative Frequency** – the proportion of observations that assume a given value

$$RF = \frac{f}{n},$$

- **Cumulative Relative Frequency** - the proportion of observations equal to or less than a given value (more on this later)

Ex. Are teens distracted by their cell phones?

Response	Frequency	Relative Frequency	Cumulative Relative Frequency
Never	289	0.39	0.39
Rarely	216	0.29	0.68
Sometimes	178	0.24	0.92
Often	60	0.08	1.00
Total	743	1.00	

Example: Mendel's Pea Plants

In one of Gregor Mendel's classic studies he bred 8,023 pea plants and recorded the color of the peas – the seeds – that each plant produced.

Pea Color	Frequency	Relative Frequency
Yellow	6022	0.751
Green	2001	0.249

The relative frequencies are almost exactly $\frac{3}{4}$ and $\frac{1}{4}$ – the 3:1 ratio Mendel's theory of inheritance predicts. A frequency table is doing real scientific work here.

Try it out: A Frequency Table by Hand

We put Pew's question to 20 students in this room: are you losing focus in class by checking your phone?

Sometimes Never Rarely Never Often
Rarely Never Sometimes Rarely Never
Sometimes Rarely Never Often Sometimes
Rarely Never Rarely Sometimes Never

Build the frequency table: frequency, relative frequency and cumulative relative frequency. Which is the modal category?

Response	f	Rel. freq.	Cum. rel. freq.
Never	7	0.35	0.35
Rarely	6	0.30	0.65
Sometimes	5	0.25	0.90
Often	2	0.10	1.00
Total	20	1.00	

Modal category: Never. Pew's 743 teens gave 0.39, 0.29, 0.24, 0.08 – 20 students landed within a few points of them.

Graphical Descriptions Of Data



Frequency Tables

Qualitative Variables

- Bar graph
- Pie Chart
- Pareto Chart

Quantitative Variables

- Dot Plot
- Stem Chart
- Histogram

Describe key features of the distribution

- Modal Category
- Shape
- Center
- Spread

Try it out: A Discrete Variable

Twenty students report how many siblings they have:

0 1 1 2 0 3 1 2 1 0
4 2 1 1 3 0 2 1 5 1

Build the frequency table. What is the modal value? Is the distribution symmetric, or skewed – and which way?

Siblings	f	Rel. freq.
0	4	0.20
1	8	0.40
2	4	0.20
3	2	0.10
4	1	0.05
5	1	0.05
Total	20	1.00

Mode = 1. Median = 1 but the mean is 1.55: the mean is pulled above the median by the long right tail, so the distribution is skewed right.

Frequency Tables for Continuous Variables

- The number of possible values is usually very large
- Convert continuous values into discrete groups (sometimes called bins):

Steps:

1. Divide the range of the variable into a set of non-overlapping intervals
2. Count the number of values that fall into each interval

Try it out: Binning a Continuous Variable

Fifteen observations of a continuous variable X:

1.23 1.38 1.76 2.60 2.65
2.79 2.89 3.12 3.15 3.18
3.65 3.77 3.87 4.12 4.41

No two values are alike, so listing them gives 15 values of frequency 1 and tells us nothing. Group them into bins of width 0.5 starting at 1.0.

Bin	f	Rel. freq.
$1.0 < X \leq 1.5$	2	0.13
$1.5 < X \leq 2.0$	1	0.07
$2.0 < X \leq 2.5$	0	0.00
$2.5 < X \leq 3.0$	4	0.27
$3.0 < X \leq 3.5$	3	0.20
$3.5 < X \leq 4.0$	3	0.20
$4.0 < X \leq 4.5$	2	0.13
Total	15	1.00

The empty bin is information: the values fall into two clusters. Next, the same procedure on real data – 272 Old Faithful eruptions.

Example: Old Faithful Eruption Times



Observation	Eruption Time	Waiting Time
1	3.600	79
2	1.800	54
3	3.333	74
4	2.283	62
5	4.533	85
6	2.883	55
7	4.700	88
8	3.600	85
9	1.950	51
10	4.350	85
11	1.833	54
⋮	⋮	
272	4.467	74

Old Faithful Eruption Times: Frequency Table

Eruption Duration (Min)	Frequency	Relative Frequency	Cumulative Relative Frequency
$1.5 \leq X < 2.0$	55	0.20	0.20
$2.0 \leq X < 2.5$	37	0.14	0.34
$2.5 \leq X < 3.0$	5	0.02	0.36
$3.0 \leq X < 3.5$	9	0.03	0.39
$3.5 \leq X < 4.0$	34	0.12	0.51
$4.0 \leq X < 4.5$	75	0.28	0.79
$4.5 \leq X < 5.0$	54	0.20	0.99
$5.0 \leq X < 5.5$	3	0.01	1.00

Visualizing Distributions of Categorical Data



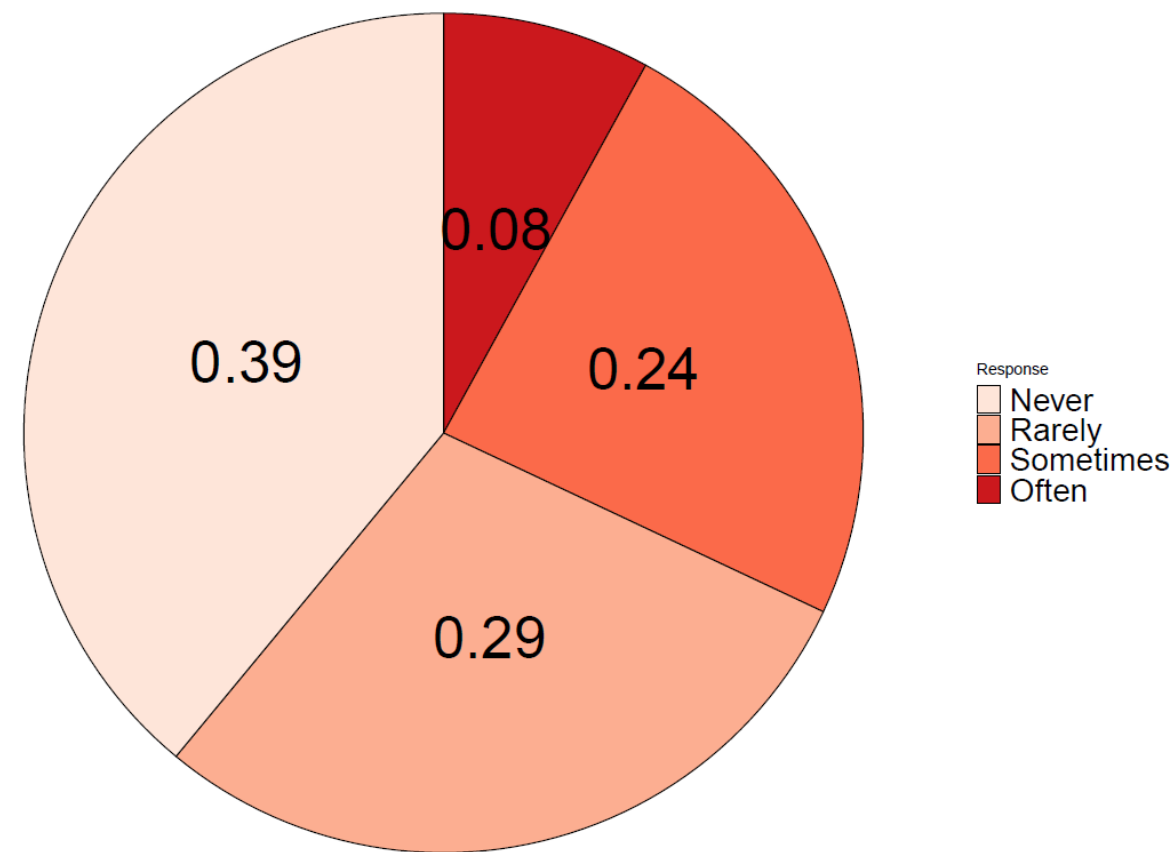
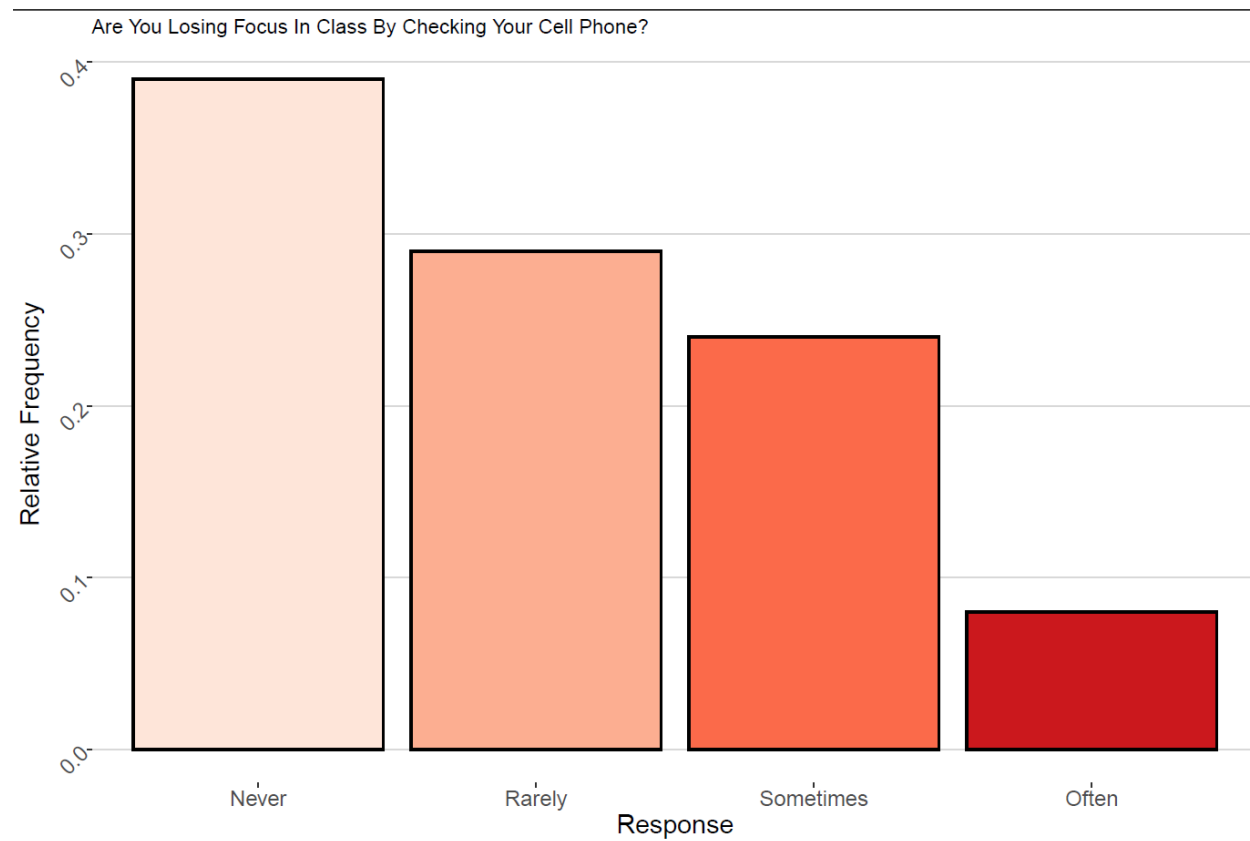
Pie Charts - a circle divided into 'slices' corresponding to each category. The size of a slice shows the proportion of observations in a category



Bar graph – displays a vertical bar for each category. The height of the bar shows the percentages of observations in the category



Pareto Chart - a bar chart with the categories ordered by decreasing frequency

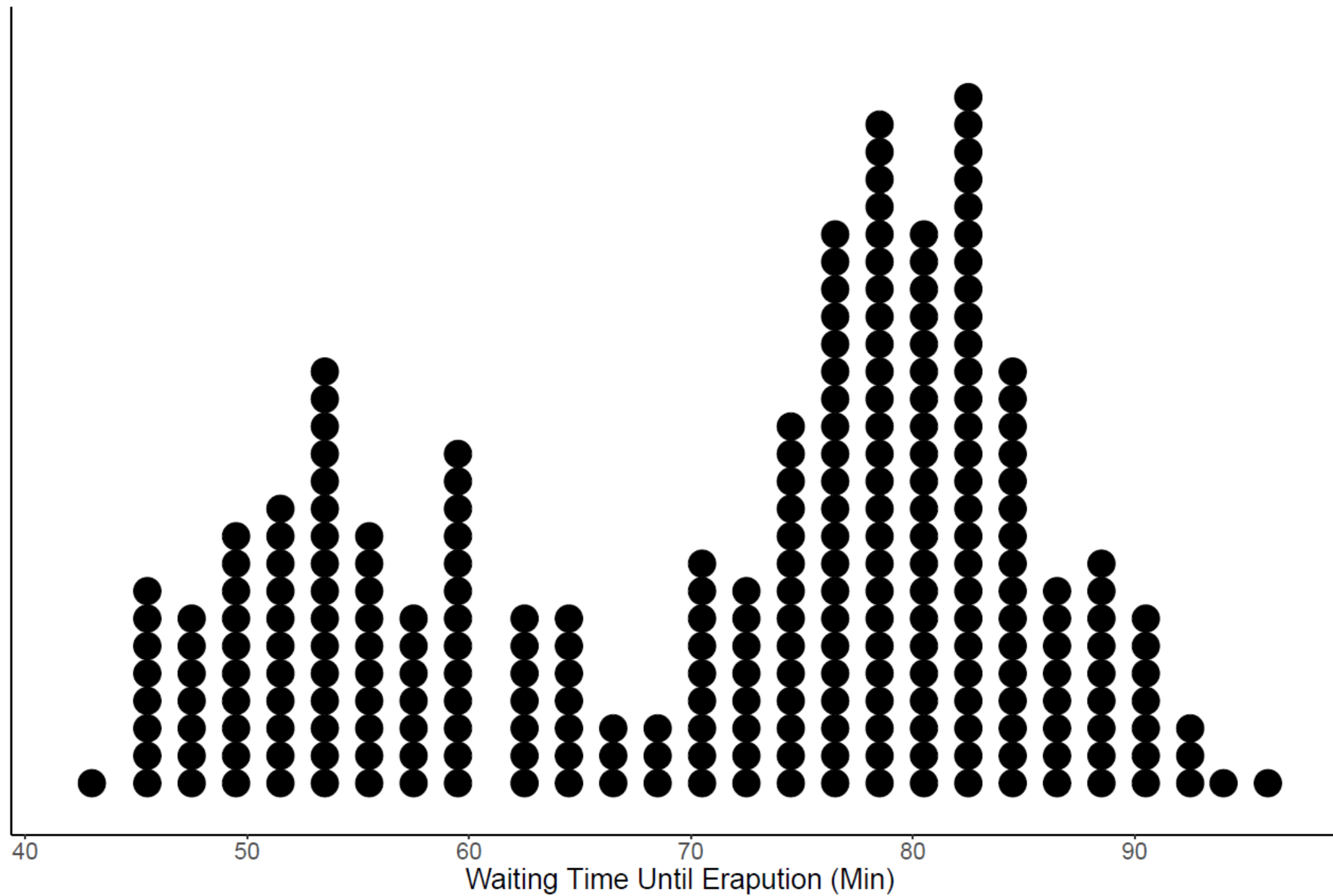


Visualizing Distributions: Quantitative Variables

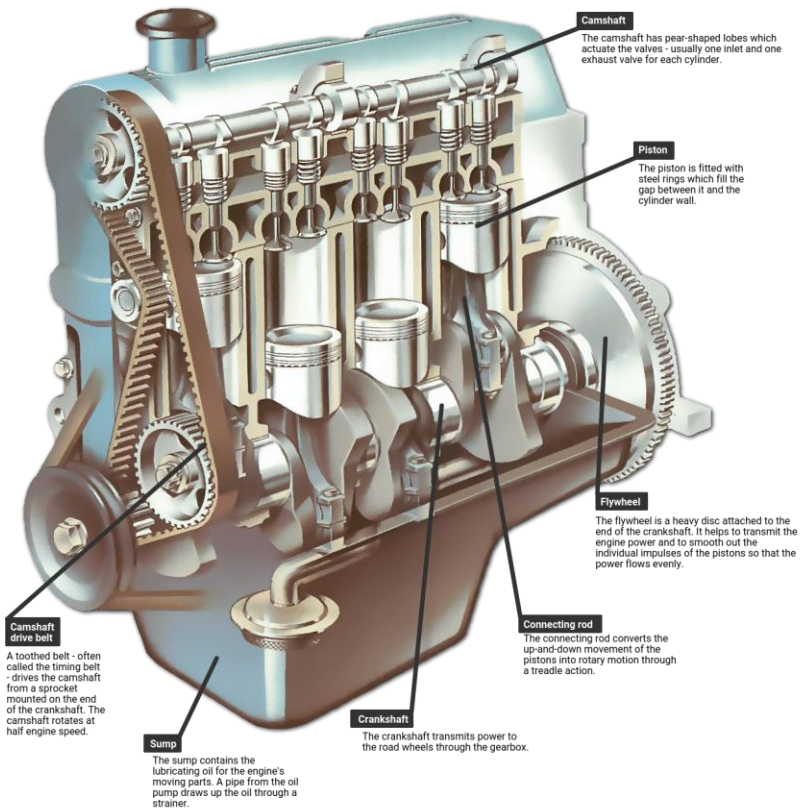
- **Dot plots** – shows a dot for each observation placed above the value for that observation

Steps to construct a dot plot

1. **Draw a horizontal line and mark the line with regular values of the variable**
2. **For each observation, place a dot above its value on the number line**
 - Works best with quantitative discrete data
 - Doesn't work well if the variable is continuous and takes on many distinct values...
 - For continuous data, the values may need to be round to the nearest tenth or integer

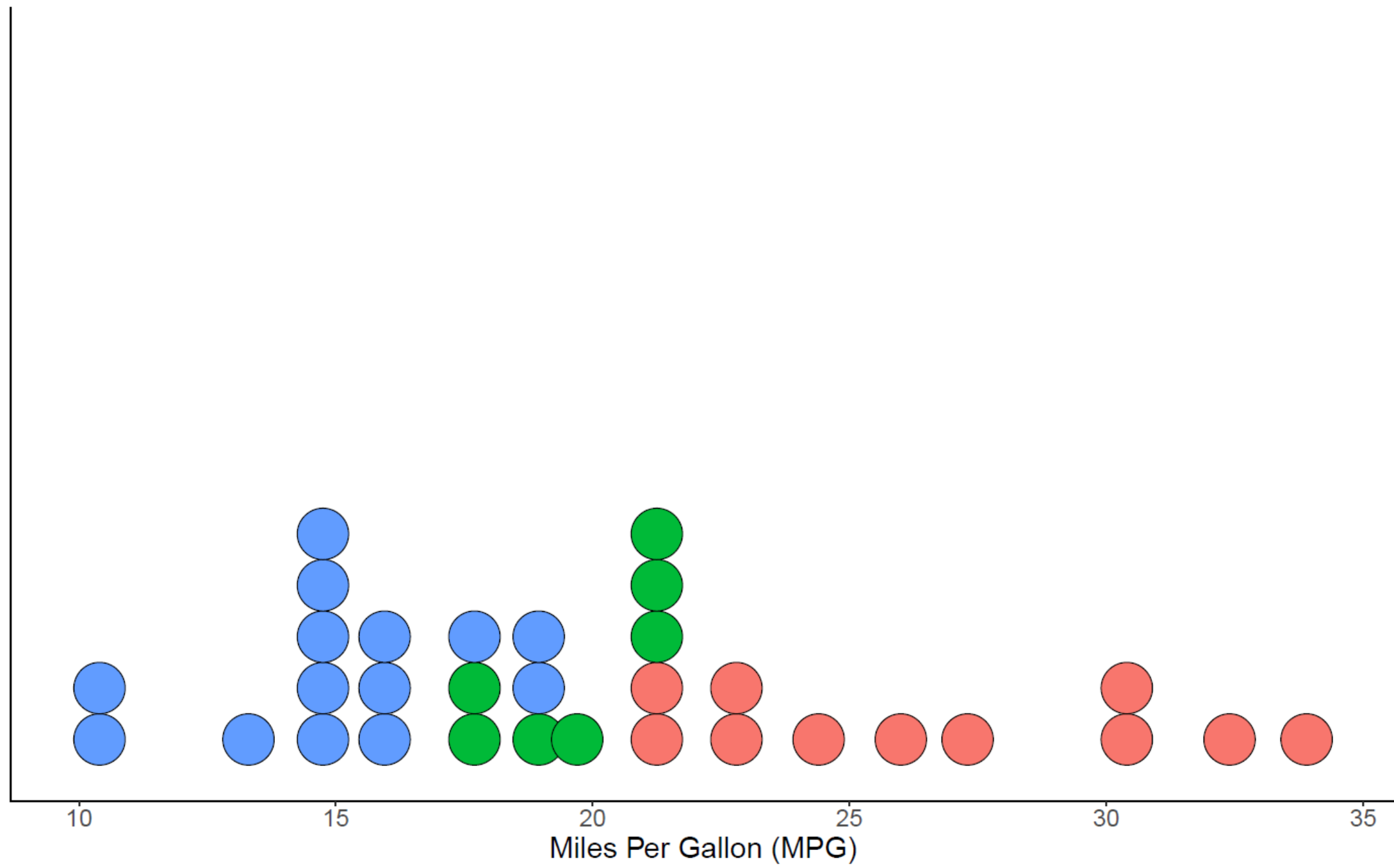


Example: MPG and Engine Cylinders



Observation	MPG	Cylinders	Model
1	21.0	6	Mazda RX4
2	21.0	6	Mazda RX4 Wag
3	22.8	4	Datsun 710
4	21.4	6	Hornet 4 Drive
5	18.7	8	Hornet Sportabout
6	18.1	6	Valiant
⋮	⋮	⋮	⋮
32	21.4	4	Volvo 142E

Number of Cylinders 4 6 8



Visualizing Distributions: Quantitative Variables

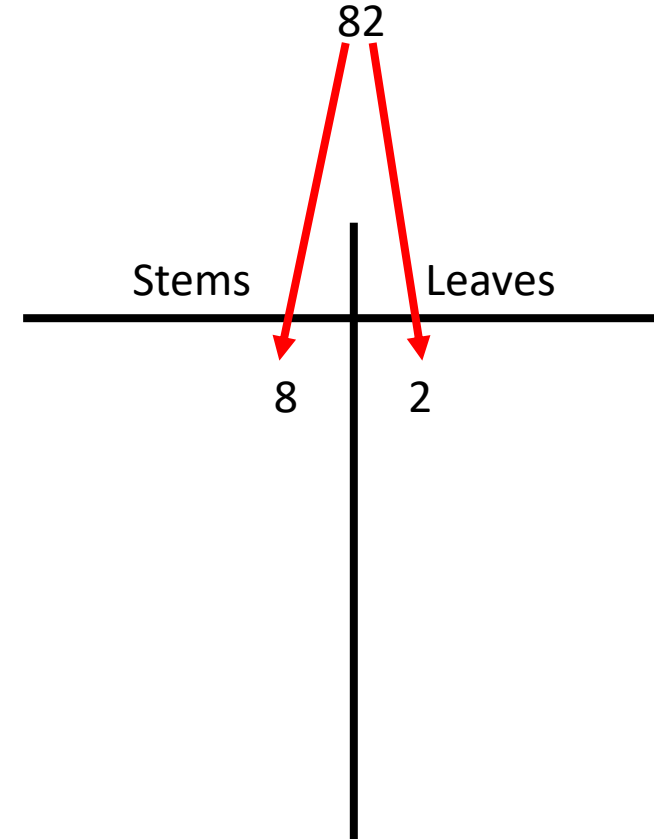
Stem and leaf plot – like a dot plot, a stem and leaf diagram also displays individual observations.

Stem – all the digits in an observation except the last digit

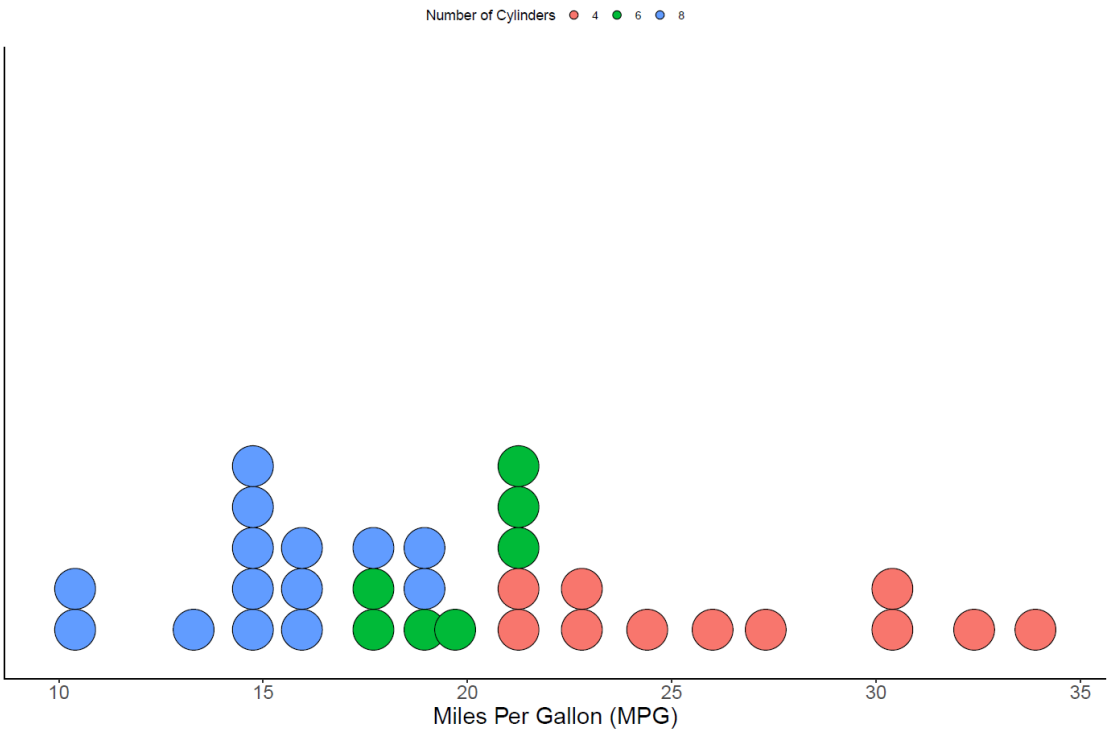
Leaf – the last digit in an observation

Steps to construct a stem and leaf plot

1. Sort the data in order from smallest to largest.
2. Place the stems in a column in increasing order
3. Place a vertical line to the right of the stems
4. To the right of the vertical line, fill in the leaves that correspond with each stem in increasing order

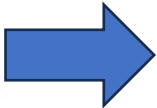


Example: MPG



Observation	MPG	Cylinders	Model
1	21.0	6	Mazda RX4
2	21.0	6	Mazda RX4 Wag
3	22.8	4	Datsun 710
4	21.4	6	Hornet 4 Drive
5	18.7	8	Hornet Sportabout
6	18.1	6	Valiant
⋮	⋮	⋮	⋮
32	21.4	4	Volvo 142E

Stems	Leaves
10	4,4
11	
12	
13	3
14	3,7
15	0,2,2,5,8
16	4
17	3,8
18	1,7
19	2,2,7
20	
21	0,0,4,4,5
22	8,8
23	
24	4
25	
26	0
27	3
28	
29	
30	4,4
31	
32	4
33	9



Stems	Leaves
10	4,4
12	3
14	3,7,0,2,2,5,8
16	4,3,8
18	1,7,2,2,7
20	0,0,4,4,5
22	8,8
24	4
26	0,3
28	
30	4,4
32	4,9

Try it out: Stem and leaf plot

Data = 4.2, 3.8, 4.6, 3.2, 2.7, 8.2, 9.1, 0.2, 1.2, 6.2

Visualizing Distributions: Quantitative Variables

Stem and leaf plots and **dot plots** are unwieldy for large n

Histogram – uses bars to portray the frequencies or relative frequencies of the possible outcomes for a quantitative variable

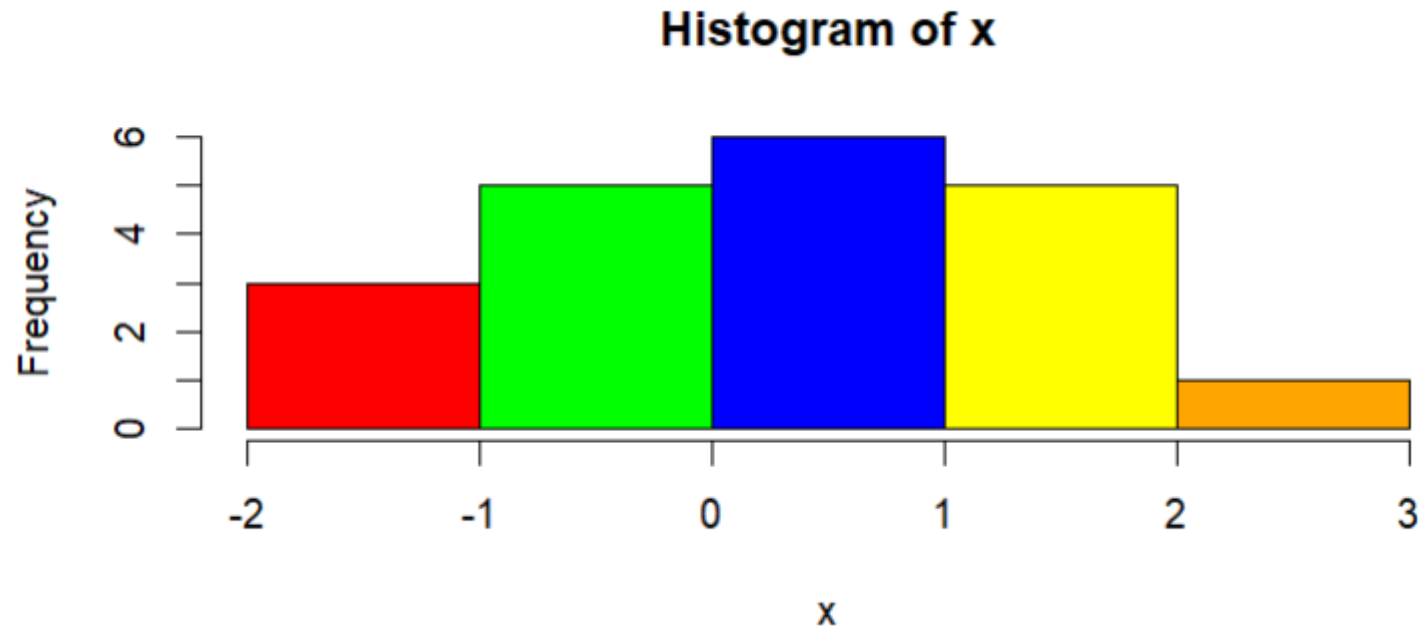
Steps to construct a histogram

1. Divide the range of the data into intervals of equal width
2. Compute the frequency of each interval (i.e construct the frequency table)
3. Label the x-axis with the values or endpoints of each interval.
4. Draw a bar over each value or interval with height equal to its frequency or relative frequency

Try it out: Histogram

Consider the following $n = 20$ observations of a continuous variable

Bin 1	Bin 2	Bin 3	Bin 4	Bin 5
-1.5, -1.2, -1.0,	-0.8, -0.7, -0.6, -0.1, -0.1,	0.1, 0.1, 0.1, 0.6, 0.6, 0.8,	1.1, 1.2, 1.3, 1.8, 1.9,	2.4



How to choose the number of Bins?

- How to choose the best number of bins is not a straightforward question and there is a lot of literature on the subject
- We can construct our histogram using a specific binwidth w or under a set number of bins k
- $w = \frac{\max x - \min x}{k}$ or $k = \frac{\max x - \min x}{w}$
- **Square root method:** $k = \text{round}(\sqrt{n})$ (A fairly safe and basic rule of thumb)
- Sturges Rule^[1]: $k = \text{round}(\log_2 n) + 1$ (not great for $n < 30$)
- Rices Rule^[2]: $k = 2\sqrt[3]{n}$

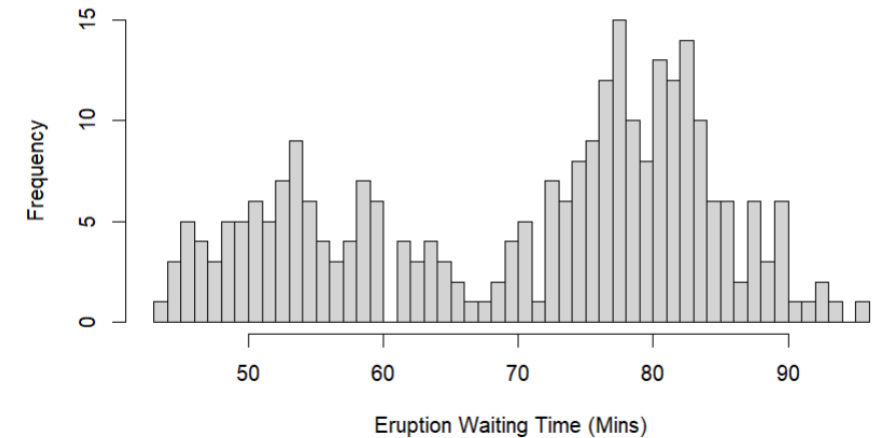
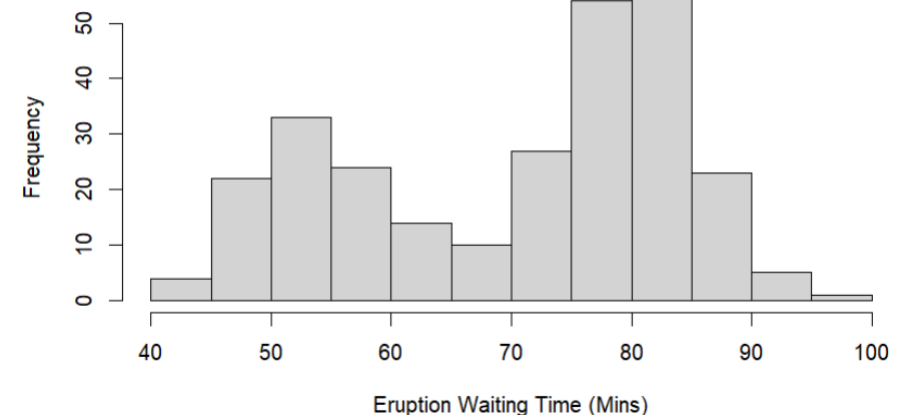
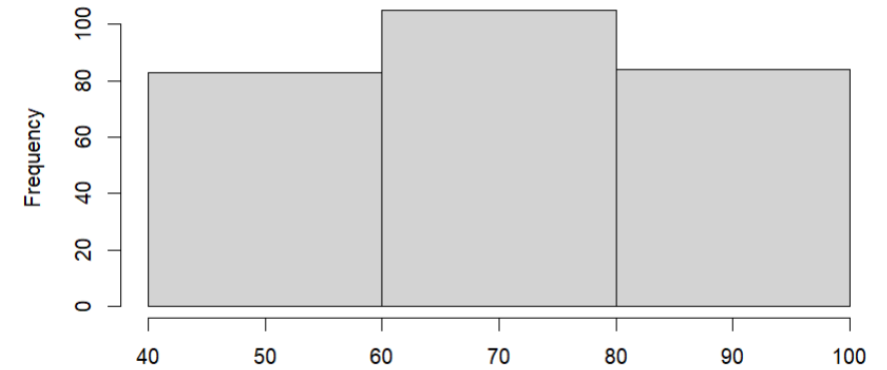
[1] Sturges, Herbert A. "The choice of a class interval." Journal of the american statistical association 21.153 (1926): 65-66.

[2] Lane, David. *Online statistics education: A multimedia course of study*. Association for the Advancement of Computing in Education (AACE), 2003. – Chapter 2 "Graphing Distributions"

Some tips

- If too few intervals are used, then the graph will be too crude
- If too many intervals are used, graph will contain many short bars and gaps.
Usually between 5 - 15 intervals are enough.
- Most plotting software will automatically choose the number of bins.
- **ALWAYS** plot the histogram to get an idea about the shape of the distribution of a quantitative variable
- Is the number of observations is small (say $n < 50$) then it's a good idea to supplement a histogram with a dot plot or stem plot

Histogram of Eruption Waiting Times



Example: Old Faithful Eruption Times

Waiting Time (Min)	Frequency	Relative Frequency	Cumulative Relative Frequency
< 50	21	0.077	0.077
50 - 60	56	0.206	0.283
60 - 70	26	0.096	0.379
70 - 80	77	0.283	0.662
80 - 90	80	0.294	0.956
> 90	12	0.044	1

Histogram of Eruption Waiting Times

